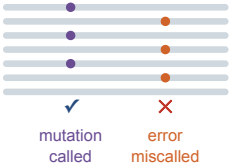
## A Where the background comes from

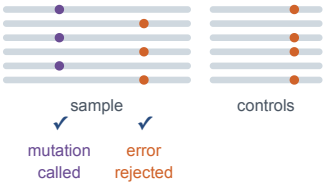
### Read-level test (e.g. LoFreq, Geneious)




Each base is judged only by its own quality score. Sequencing errors that recur at one position (PCR, homopolymers, paralog reads) pass this test and are wrongly counted as SHM mutations, and reads from one animal are treated as independent replicates.

✓ correct decision ✕ wrong decision

### Control-calibrated test (SHMtrace)




Error rates per substitution, strand and base quality are measured in matched AID-null libraries from the same run. A call needs support beyond this background (FDR ≤ 0.05), and groups are compared with one value per animal.

## B Inputs and read preparation



**Nanopore**
full-length amplicons



**Sanger**
.ab1 traces + sample sheet



**Illumina**
paired reads, merged



**UMI-tagged**
one consensus per molecule

### Read preparation

1. **Length and expected-error gates**
   over-length reads and reads averaging > 0.01 errors per base
2. **Competitive alignment**
   minimap2 against target and paralog references; chimeras out
3. **Paralog-allele filter**
   reads with ≥ 10 paralog-specific bases removed
4. **Matched depth**
   random draw of the same number of usable reads per library

## C Calling and quality checks

**1 Control error model**
per substitution, strand and base quality; a margin chosen by leave-one-control-out

→

**2 Test and FDR**
support against expected errors; FDR ≤ 0.05 set from the controls; each control called vs the others

→

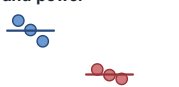
**3 Germline and window**
sites at ≥ 50% removed; calls restricted to the covered window; indels optional

→

**4 Automatic checks**
spectrum (C→T/G→A share), positional clustering, 5-mer targeting index, Sanger bookkeeping flags

## D Outputs

**Animal-level tests and power**
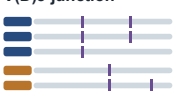


NB, Welch, permutation

**Single-read estimate**



observed minus control background

**Clones from the V(D)J junction**



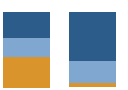
reads grouped by junction; each clone counted once

**Mutation trees**



full-length reads of one clone

**Repair-route indices**



G:C Ts / Tv, A:T, WA/TW, decomposition

**Reports, app and command line**



HTML report, tables, project files